



OPEN Ce:LYSO, from scintillator to solid-state lighting as a blue luminescent concentrator

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Cerium-doped lutetium-yttrium oxyorthosilicate (Ce:LYSO) is a well-known single crystal scintillator used in medical imaging and security scanners. Recent development of high power UV LED, matching its absorption band, questions the possibility to use Ce:LYSO in a new way: as LED-pumped solid-state light source. Since Ce:LYSO is available in large size crystals, we investigate its potential as a luminescent concentrator. This paper reports an extensive study of the performance in close relation to the spectroscopic properties of this crystal. It gives the reasons why the Ce:LYSO crystal tested in this study is less efficient than Ce:YAG for luminescent concentration: limited quantum efficiency and high losses coming from self-absorption and from excited-state absorption are playing key roles. However, we demonstrate that a Ce:LYSO luminescent concentrator is an innovative source for solid-state lighting. Pumped by a peak power of 3400 W in quasi-continuous wave regime (40 μ s, 10 Hz), a rectangular ($1 \times 22 \times 105$ mm³) Ce:LYSO crystal delivers a broadband spectrum (60 nm FWHM) centered at 430 nm. At full output aperture (20×1 mm²), it emits a peak power of 116 W. On a squared output surface (1×1 mm²) it emits 16 W corresponding to a brightness of 509 W cm⁻² sr⁻¹. This combination of spectrum power and brightness is higher than blue LEDs and opens perspectives for Ce:LYSO in the field of illumination namely for imaging.

Among the materials for solid-state lighting, Ce:YAG is certainly the crystal with the longest history. Its spectroscopy has been extensively studied in the 1970s^{1–3}. It started to be used as scintillator in the 1990s⁴ and it is broadly used for beta radiation or X-ray counting, electron or X-ray imaging screens. The first use as solid-state lighting is reported in 1997⁵, with a wavelength conversion by absorption of a blue GaN LED and an emission in the yellow-orange. Next Ce:YAG has been used as a light conversion phosphor in multiple configurations: powders⁶, ceramics⁷, single crystals pumped by blue lasers diodes⁸, and more recently LED-pumped Ce:YAG luminescent concentrators^{9–12}.

During the last 30 years, many materials were selected and developed for scintillator applications¹³. As this field requires large size crystals with good optical quality, it is interesting to investigate if any other crystal dedicated to scintillation have a potential for solid-state lighting and for new luminescent concentrators. Among them, Ce:LYSO (Ce:Lu_{2(1-x)}Y_{2x}SiO₅) is known to be an excellent scintillator for positron emission tomography (PET-scan), calorimetry in high energy physics, or security scanners. Investigations on this crystal started 20 years ago^{14–16}. Its crystal growth process is now well mastered. Ce:LYSO is commercially available in large dimensions (up to 100 mm) with excellent optical quality for scintillation applications. Ce:LYSO absorption band around 350 nm matches quite well the emission spectrum of high power UV LEDs that have been recently developed for optical adhesive curing, sterilisation or bio/medical treatments. Moreover, Ce:LYSO emits a broadband spectrum that complete the Ce:YAG spectrum in the bluer range (Fig. 1). Therefore, Ce:LYSO appears to be a good candidate for a LED-pumped luminescent concentrator emitting in the blue.

The purpose of this work is to investigate its potential through an extensive experimental study supported by an analytical model. Firstly, we briefly investigate the spectroscopy of Ce:LYSO under optical pumping in the UV. Then, we present the experimental setup of a LED-pumped Ce:LYSO luminescent concentrator and the performance in terms of peak power and brightness. Afterward, we confront the experimental results with an analytical model showing the origin of the limitations in Ce:LYSO performance. Finally, we compare this new light source to other blue sources and conclude about the potential of LED-pumped Ce:LYSO luminescent concentrators.

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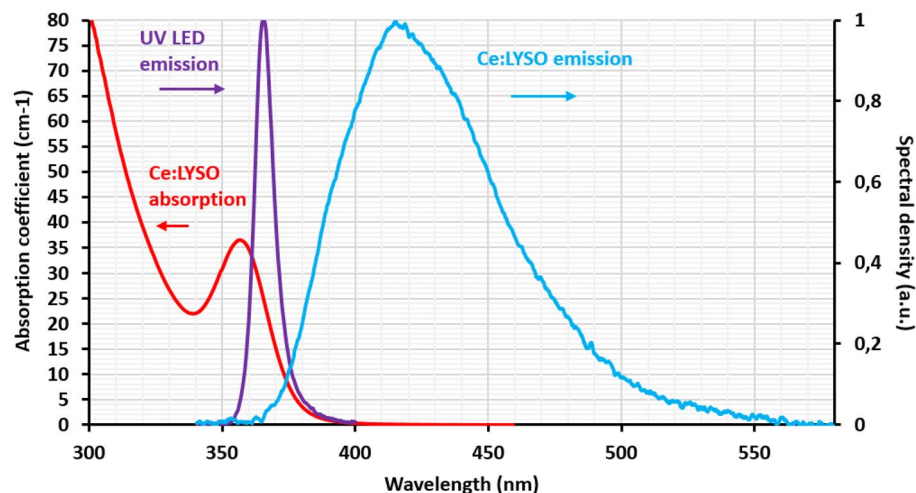


Figure 1. b-cut Ce:LYSO absorption and emission spectra compared to 365 nm UV LED emission spectrum (Nichia, NVCUQ096A-U365). None of the light sources are polarized.

Spectroscopic investigations

To our knowledge, this is the first time that a Ce:LYSO is optically pumped in the UV. Samples used in this work are large single crystals from Crytur.cz company. Emission and absorption spectroscopy are performed using a small sample of $10 \times 2 \times 1$ mm³. Absorption measurements are carried out with a UV/Vis/NIR spectrophotometer (Varian Cary 6000). The spectra are reported in Fig. 1. The crystal orientation has been identified with Laue X-ray diffraction method. Absorption and emission are measured for polarizations parallel to the two different dielectric axes perpendicular to b monoclinic axis. No significant differences have been observed between the different polarizations, as reported previously¹⁷. From the absorption measurements, we deduced that the Ce³⁺ concentration is 0.05% in our samples.

Quantum yields efficiencies were assessed using C9920-02G PL-QY measurement system from Hamamatsu consisting in a 150 W monochromatized Xe lamp, an integrating sphere (Spectralon coating, $\varnothing = 8.4$ cm) and a high sensibility CCD camera. We experimentally obtained a quantum efficiency of $\eta_r = 0.51$ for a pump wavelength at 365 nm, corresponding to the operating central wavelength of the LED that will be used for pumping a Ce:LYSO luminescent concentrator. With the same experimental setup and for a sake of comparison, we also measure a quantum efficiency for a commercial Ce:YAG (PhosphorTech) pumped at 460 nm (its excitation wavelength). This value for Ce:YAG is significantly higher with $\eta_r = 0.95$.

Two hypotheses can be put forward to understand this relatively low quantum efficiency. A first possible explanation comes from the position of the 5d bands within the Ce:LYSO bandgap. The lowest 5d level (namely 5d¹ level) of Ce³⁺ is located inside the bandgap, ensuring a good emission, but the second and third lowest 5d levels are located within the conduction band. Therefore a low quantum efficiency could be related to a quenching process in the Ce:LYSO host in relation with the thermally stimulated ionization process from the lowest 5d¹ level to the conduction band. This process has been widely reported in the garnets for instance comparing Ce:YAG and Ce:YAGG (Ce:Y₃Al₂Ga₃O₁₂)¹⁸. Similar situation has also been observed in various oxides compounds¹⁹. A second explanation comes from the absorption spectrum of our Ce:LYSO samples as presented in Fig. 1. Indeed, the rising absorption front below 320 nm can be related to Ce⁴⁺ species. Tetravalent cerium is actually required to improve scintillation light yield, as reported in references^{20,21}, and its presence is typically promoted by the crystal growth atmosphere containing a little amount of dioxygen. However, Ce⁴⁺ species can absorb the pump light in the place of Ce³⁺ and hence reduce the emission in the blue. To corroborate the effect of the tetravalent cerium on the quantum efficiency, a thermal annealing was performed under reducing atmosphere on a small sample, and this indeed increase the quantum yield. However, we observed that the optical quality of the sample was affected. Thermal annealed large pieces of Ce:LYSO would certainly require more investigations that are out of the purpose of this work which presents the use of commercially available Ce:LYSO crystals optimized for scintillators.

LYSO structure presents also two distinct crystallographic sites for Ce³⁺ ions, one in coordination 6 and the other in coordination 7 with a more distorted polyhedron. The two resulting Ce³⁺ centers, usually referred as Ce1 and Ce2, have different optical properties and hence different quantum yields. Ce1 has the highest quantum yield, measured at 78% at T = 14 K²², with main emission peaks at 393 and 427 nm²³, whereas Ce2 has weaker quantum yield although it is the most abundant²⁴ with main emission peak around 460 nm. As Ce1 and Ce2 centers notably present excitation bands at 356 and 376 nm respectively²³, both are excited when performing room temperature excitation at 365 nm, so that the resulting mean quantum yield is an average between the two centers contributions.

As the overlap between the absorption spectrum and the emission spectrum is significant (Fig. 1), we investigate the emission spectrum versus different propagation lengths in the Ce:LYSO. For this purpose, we use a frequency tripled Nd:YAG as pump source (emission at 355 nm) and move its 5 mm diameter pump spot along our 315-mm-long Ce:LYSO sample. This sample is composed of 3 parallelepiped Ce:LYSO single all-faces-polished

crystals (each of them having dimensions of $1 \times 22 \times 105 \text{ mm}^3$). The parallelepipeds are bonded together with a UV-curing optical adhesive (refractive index of $n = 1.56$) to minimize reflections at the interface between bonded crystals. Ce:LYSO crystal absorbs the UV light at 355 nm and emits a blue light localized in the pump spot. A part of this converted light propagates in the sample by total internal reflections and reaches the output surface where the emitting spectrum is collected with an integrating sphere connected to a spectrometer. The different output spectra shown in Fig. 2 are normalized assuming that self-absorption is negligible for wavelengths larger than 450 nm and that the spectra are then similar in the reddest regions. Figure 2 shows a significant evolution of the emission spectrum versus the propagation length inside the crystal. This can be related to self-absorption losses for light guided inside the crystal by total internal reflections (TIR). Indeed, after each self-absorption-re-emission process, the light is redistributed in all directions and part of it can again escape through the faces of the crystal. The knowledge of these losses is thus crucial for luminescent concentrators.

In order to evaluate these losses, one can consider the propagation over a distance l as a first example. The spectral densities to be compared are $I_0(\lambda)$ and $I_l(\lambda)$ (Fig. 2), measured when the pump spot is close to the output surface (at $z = 0$) and at a distance $z = l$, respectively. The photons coming from the pump spot at $z = l$ propagate in the crystal towards the output face by TIR in directions included in a cone (called the output escape cone) with an apex angle $\theta_{TIR} = \sin^{-1}(\frac{1}{n})$, n being the refractive index of the crystal. In²² we have shown that the average propagation distance is $d = Kl$ with

$$K = \frac{-\ln(\cos \theta_{TIR})}{(1 - \cos \theta_{TIR})} \quad (1)$$

For each wavelength, one can deduce a loss coefficient caused by self-absorption given by:

$$\alpha(\lambda) = -\frac{1}{Kl} \ln \left(\frac{I_l(\lambda)}{I_0(\lambda)} \right) \quad (2)$$

Average self-absorption losses coefficient α_l can be found by integrating all over the output spectrum:

$$\alpha_l = \frac{\int_{390\text{nm}}^{550\text{nm}} \alpha(\lambda) I_l(\lambda) d\lambda}{\int_{390\text{nm}}^{550\text{nm}} I_l(\lambda) d\lambda} \quad (3)$$

With the data recorded on the experiment described on Fig. 2, we found $\alpha_l = 1.7 \times 10^{-2} \text{ cm}^{-1}$.

For comparison, we propagate a HeNe laser (at 633 nm) in the Ce:LYSO crystal, along 105 mm (without reflexions on faces). We found a loss coefficient of $\alpha_0 = 10^{-3} \text{ cm}^{-1}$ showing the excellent optical quality of our sample. This means that self-absorption losses are much more significant than losses due to crystal imperfections, the assumption of normalization at long wavelengths is thus validated.

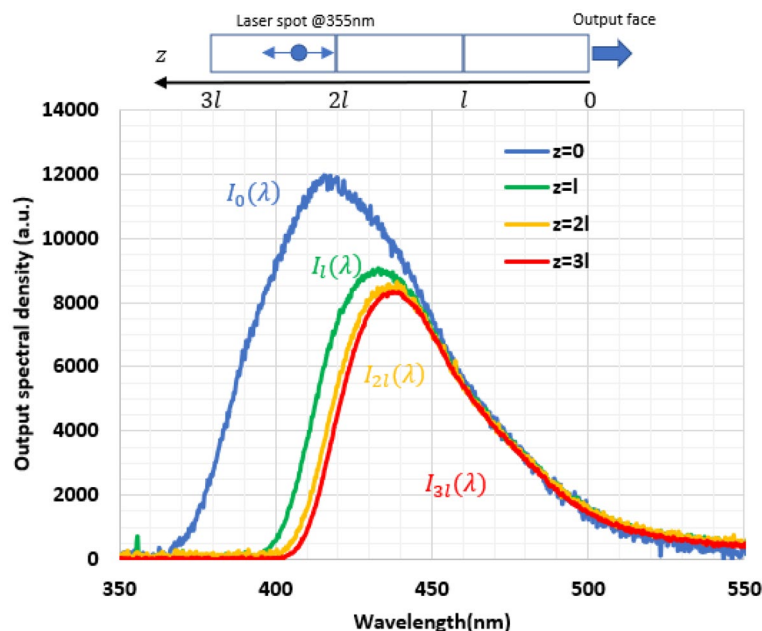


Figure 2. Ce:LYSO emission spectra recorded at the output face and pumped by a frequency tripled Nd:YAG laser emitting at 355 nm. $l = 105 \text{ mm}$. The sample dimensions are $1 \times 22 \times 315 \text{ mm}^3$ with $l = 105 \text{ mm}$. Spectra are normalized for wavelengths larger than 450 nm.

LED-pumped Ce:LYSO luminescent concentrator: experimental setup

The experimental setup of the LED-pumped luminescent concentrator is presented in Fig. 3. The Ce:LYSO concentrator is cut as a parallelepiped, whose length, width, and thickness are $l = 105$ mm, $w = 22$ mm and $t = 1$ mm, respectively. The six faces of the Ce:LYSO are optically polished with a scratch/dig of 60/40. The parallelism between the two large faces " $w.l$ " is measured to be 3.4λ (peak to valley) at 633 nm, by interferometry.

The LC is pumped on both sides, across its two largest ($w.l$) surfaces by 768 UV LEDs arranged in 8 arrays (Nichia, NVCUQ096A-U365), brazed on ceramic printed circuit boards (PCB). The PCBs are mounted on aluminum blocks with water cooling for temperature control. Each array is composed of 96 LEDs divided in 12 lines of 8 LEDs. The 8 arrays are represented on Fig. 3. They are close to the large faces of the Ce:LYSO (typically at 1 mm). Each PCB can provide an optical power of 123 W in continuous wave (CW) for a nominal drive current of 8 A corresponding to a voltage of 46.3 V. The LED emission spectrum is centered at 365.5 nm and has spectral width of 10 nm at full width at half maximum (Fig. 1). In order to increase the output peak power, the LEDs are activated in quasi continuous wave (QCW), during 40 μ s at 10 Hz. At this very low duty cycle (0.04%), it is possible to increase the drive current up to 30 A, corresponding to a peak power of 426 W per PCB. The total peak power available is then 3405 W.

The spectrum of the LEDs at maximum current is shown on Fig. 1. The spectral overlap of UV LEDs emission with Ce:LYSO concentrator absorption band gives an average absorption coefficient of $\alpha_p = 17.7 \text{ cm}^{-1}$ (see Table 1).

In order to preserve total internal reflections in the concentrator, Ce:LYSO is maintained by 4 small pins. This means that the Ce:LYSO is quasi exclusively radiatively cooled. With a duty cycle of 0.04%, the average pump power reaches a maximum of 1 W. Assuming that a maximum of 50% of the power is converted into heat (corresponding to the quantum efficiency), we calculated a temperature increase of 7 °C. The quantum efficiency of the Ce:LYSO is temperature dependent¹⁵ but performance are only slightly affected by this small temperature increase.

The output face of the system is one of the small faces (surface $S = w.t$). In order to increase the output power and the brightness, mirrors (with the E02 coating from Thorlabs: HR 400–750 nm) can be positioned close to the $w.t$ faces. A so-called "back" mirror (Fig. 3) put on the opposite face to the output face then allows to recycle the rays emitted in the backward direction compared to the output face. This is a classical configuration already

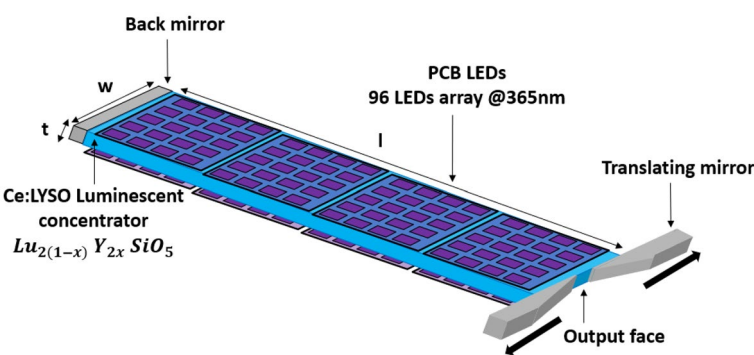


Figure 3. Experimental setup of the LED pumped Ce:LYSO luminescent concentrator. The b-cut Ce:LYSO dimensions are $t = 1$ mm, $w = 22$ mm and $l = 105$ mm. Beveled translating mirrors on the output face are used to adjust the output surface and the brightness.

Parameter description	Ce:YAG (0.11% doping)	Ce:LYSO (this work) (0.05% doping)	Comments for Ce:LYSO
Refractive index	$n = 1.82$	$n = 1.83$	
Pump wavelength (peak)	$\lambda_p = 440 \text{ nm}$	$\lambda_p = 365 \text{ nm}$	Mean wavelength of the LED
Absorption coefficient	$\alpha_p = 18 \text{ cm}^{-1}$	$\alpha_p = 17.7 \text{ cm}^{-1}$	Averaged over the LED spectrum
Emitted wavelength	$\lambda_{em} = 550 \text{ nm}$	$\lambda_{em} = 450 \text{ nm}$	Mean wavelength of the emitted spectrum
Lifetime of the emitting level	$\tau = 70 \text{ ns}^4$	$\tau = 41 \text{ ns}$	From measurement, in agreement to ¹⁴
Luminescence quantum efficiency	$\eta_r = 0.95$	$\eta_r = 0.51$	
Diffusion loss coefficient	$7 \times 10^{-4} \text{ cm}^{-125}$	10^{-3} cm^{-1}	Measured at 633 nm
Self-absorption loss coefficient	$1.6 \times 10^{-3} \text{ cm}^{-1}^{25}$	$4 \times 10^{-3} \text{ cm}^{-1}$	Deduced from measurements for propagation distances larger than $z = 20 \text{ cm}$, correspond to α_{3l}
Excited state absorption cross section	$1.2 \times 10^{-17} \text{ cm}^2$	$2.5 \times 10^{-17} \text{ cm}^2$	Fit parameter adjusted for P_{3D} at a pump peak power of 3405 W

Table 1. Physical parameters of Ce:LYSO as luminescent concentrator compared with Ce:YAG. For the Ce:YAG, data come from¹² excepted when it is mentioned in the table.

reported in the literature^{9–11}. In the following, this configuration refers as a "2D" configuration because the light emerging on the output face has been confined in the two lateral directions of space by total internal reflections.

In order to increase furthermore the output brightness, we can add two mirrors in order to reduce the output face size as shown in the Fig. 3²⁵. In this configuration, the light is also confined in the longitudinal direction leading then to a confinement in the three dimensions of space, hence the name "3D" configuration. By translating the mirrors in a plane parallel to the $w.t$ plane (Fig. 3), the output aperture surface, denoted s , can be varied from $s = S = w.t = 22 \times 1 \text{ mm}^2$ to $s = 0.55 \times 1 \text{ mm}^2$ corresponding to an aperture ratio $r = S/s$ varying from $r = 1$ to $r = 40$. These mirrors are beveled to avoid light aperturing in the case of very small apertures.

The next parts present results and analysis in the "2D" configuration (i.e. without mirrors on the output face) and in the "3D" configuration (i.e. with mirrors on the output face).

Performance and analysis in the "2D" configuration

The output peak power is measured with an integrating sphere coupled to a spectrometer (LCS-100 from Lab-sphere). As the output beam is Lambertian and strongly divergent, the entrance of the integrating sphere is positioned very close to the output surface for accurate measurements. The evolution of the output peak power versus the pump peak power is reported in Fig. 4, with and without the "back" mirror. The maximum peak power is 116 W for a pump peak power of 3405 W corresponding to a brightness of $167 \text{ W cm}^{-2} \text{ sr}^{-1}$ and an optical efficiency of 3.4%. In comparison, a Ce:YAG material in a similar setup provides an optical efficiency of 7.7%¹².

In order to understand the origin of the limited performance, we modeled the output power as follows. Self-absorption losses are taken in account by separating the emission in two contributions P_1 and P_2 , with respect to the propagation distance in the crystal.

P_1 is the power related to "one-way" photons that are emitted in the output escape cone of the $w.t$ output face.

P_2 is the power related to photons emitted in the backward emission cone and that propagate towards the output face after a reflection on the back mirror.

The expression of P_1 is found by integration between $z = 0$ and $z = l$ of all the emission in the output cone.

$$P_1 = T_{LED} \cdot \eta_r \cdot \frac{\lambda_p}{\lambda_{em}} P_{pump} (1 - e^{-\alpha_p}) \frac{1 - \cos \theta_{TIR}}{2} \frac{1}{\alpha_1 K l} (1 - e^{-\alpha_1 K l}) \quad (4)$$

where T_{LED} is the transmission of the $w.l$ face for the LED: As the LED emission cone is 30° , this quantity can be approximated by Fresnel transmission at normal incidence ($T_{LED} = 0.91$), λ_p is the average pump wavelength (367 nm), λ_{em} is the average emitted wavelength (455 nm), P_{pump} is the peak pump power (from 0 to 3405 W), α_1 is the loss propagation coefficient for these photons propagating from 0 to l in the concentrator. The loss has 3 contributions:

$$\alpha_1 = \alpha_0 + \alpha_l + \alpha_{ESA} \quad (5)$$

where α_0 corresponds to the losses of the concentrator including the crystal quality, including diffusion loss coefficient and the polishing quality ($\alpha_0 = 4 \times 10^{-3} \text{ cm}^{-1}$), α_l corresponds to the self-absorption losses ($\alpha_l = 1.7 \times 10^{-2} \text{ cm}^{-1}$) and is calculated using spectral absorption data presented in Fig. 2, α_{ESA} corresponds to the losses induced by excited state absorption $\alpha_{ESA} = \sigma_{ESA} n_2$, n_2 being the population density of the upper state level. As in¹², we neglect the decrease of n_2 , caused by ESA because of the low pump power density of LEDs. Therefore, the relation between n_2 and the pump power P_{pump} is:

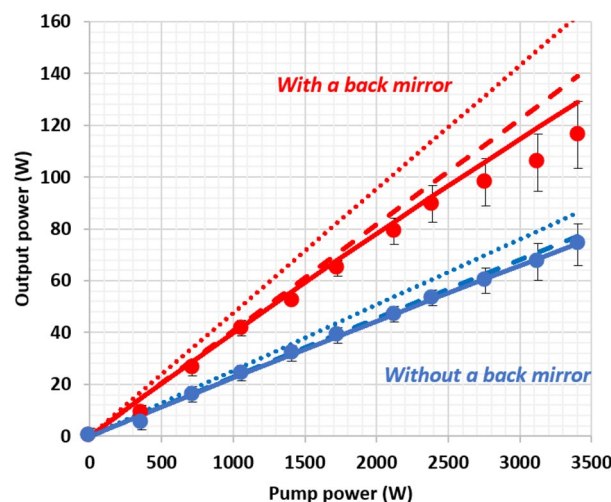


Figure 4. Peak power emitted by the LC versus the pump peak power with and without "back" mirror. Dots are experimental measurements. Plains curves are simulations including excited state absorption (ESA) and self-absorption losses. Dashed lines are simulations taking into account the self-absorption losses but not the ESA. Dot lines are simulations without taking into account ESA nor self-absorption.

$$n_2 = \tau \frac{P_{\text{pump}} T_{\text{LED}}}{l w t} \frac{\lambda_p}{h c} (1 - e^{-t \alpha_p}) \quad (6)$$

where τ is the lifetime of the upper level ($\tau = 41$ ns).

The expression for P_2 is similar to the expression for P_1 but with an extra reflection on the opposite face and an additional propagation between l and $2l$ in order to reach the output face.

In the case of a back mirror, the expression of P_2 is:

$$P_2 = R P_1 e^{-\alpha_2 K l} \quad (7)$$

where

$$\alpha_2 = \alpha_0 + \alpha_{2l} + \alpha_{\text{ESA}}$$

α_{2l} corresponds to the self-absorption losses for a propagation between l and $2l$. It is calculated similarly as α_l by considering the spectral density $I_l(\lambda)$ and $I_{2l}(\lambda)$ presented on Fig. 2 ($\alpha_{2l} = 9 \times 10^{-3} \text{ cm}^{-1}$).

Without the back mirror, a Fresnel reflection on the opposite face must be considered, the expression of P_2 is therefore:

$$P_2 = (1 - T_{\text{Fresnel}}) P_1 e^{-\alpha_2 K l} \quad (8)$$

where T_{Fresnel} is the Fresnel transmission of the $w.t$ interfaces taking all directions in the output escape cone and all polarizations in account ($T_{\text{Fresnel}} = 0.86$).

For the calculation of the output power, we consider the light transmitted by the output $w.t$ interface (related to T_{Fresnel}) and the light reflected by this interface coming back after a reflection on the opposite face. Multiple reflection light can be simulated as in²⁵. It corresponds to propagation lengths in the concentrator larger than $2l$.

Therefore, the output power with a back mirror having a reflectivity R is:

$$P_{2D\text{mirror}} = (P_1 + P_2) \frac{T_{\text{Fresnel}}}{1 - e^{-2\alpha_3 K l} R (1 - T_{\text{Fresnel}})} \quad (9)$$

where α_3 correspond to losses for photons propagating distances larger than $2l$ in the concentrator $\alpha_3 = \alpha_0 + \alpha_{3l} + \alpha_{\text{ESA}}$

For those photons, self-absorption losses are estimated from our measurements presented in Fig. 2. Considering the spectral density $I_{2l}(\lambda)$ and $I_{3l}(\lambda)$ a loss coefficient namely α_{3l} can be deduced ($\alpha_{3l} = 4 \times 10^{-3} \text{ cm}^{-1}$). As the spectral shape evolves only slightly between $2l$ and $3l$, we can suppose that this value can also be used for longer propagation distances than $3l$.

Without the back mirror, the output power is found by replacing R by $1 - T_{\text{Fresnel}}$:

$$P_{2D} = (P_1 + P_2) \frac{T_{\text{Fresnel}}}{1 - e^{-2\alpha_3 K l} (1 - T_{\text{Fresnel}})^2} \quad (10)$$

The simulations are presented in Fig. 4. In order to separate the different limitations, we calculate also the output power with and without excited-state absorption (ESA) and self-absorption. Figure 4 shows that self-absorption has a stronger influence than ESA. ESA effect is more visible with the back mirror since longer propagation distances render this configuration more sensitive to losses.

The simulations are adjusted to the experimental data thanks to three loss parameters that have not been measured: the effective reflectivity R of the back mirror (taking the large divergence of the output beam in account), the cross section of the excited-state absorption and the effective loss coefficient α_0 taking in account crystal imperfections (diffusion measured to be 10^{-3} cm^{-1} and polishing quality). For a better accuracy, those parameters are adjusted by fitting the output power obtained in the "3D configuration" (see section "Performance and analysis in the 3D configuration") which is much more sensitive to losses because of long propagation distances inside the concentrator. Despite the indirect adjustment, Fig. 4 shows that the 2D configuration simulations agree with experimental value s .

Performance and analysis in the "3D" configuration

In this section, we have measured the output power by translating the mirrors along the output face (see Fig. 2). At full aperture (aperture ratio $r = 1$) and full pump peak power (3405 W), we measure an output power of $P_{3Dr=1} = P_{2D\text{mirror}} = 116$ W. At $r = 22$ corresponding to a squared output surface of 1 mm^2 , the output peak power decreases to $P_{3Dr=22} = 16$ W. This effect is mainly due to the propagation losses inside the Ce:LYSO. However, the brightness ($\text{W cm}^{-2} \text{ sr}^{-1}$) is increased by a factor 4. Light recycling in this particular "3D" configuration with $r = 22$ has two assets: first it increases the brightness, and second it symmetrizes the output surface in a square shape. Figure 5 gives the output brightness versus the aperture ratio for two different pump powers.

Assuming that the output emission is Lambertian, the brightness of the output area can be deduced:

$$B_{3D} = \frac{P_{3D} \cdot r}{\pi \cdot w \cdot t} \quad (11)$$

where P_{3D} is the power emitted by the reduced output surface s and r is the aperture ratio $r = S/s$.

The output power P_{3D} can be expressed by light recycling of P_1 and P_2 after every bounce on the mirrors (having the same reflectivity R)²⁵:

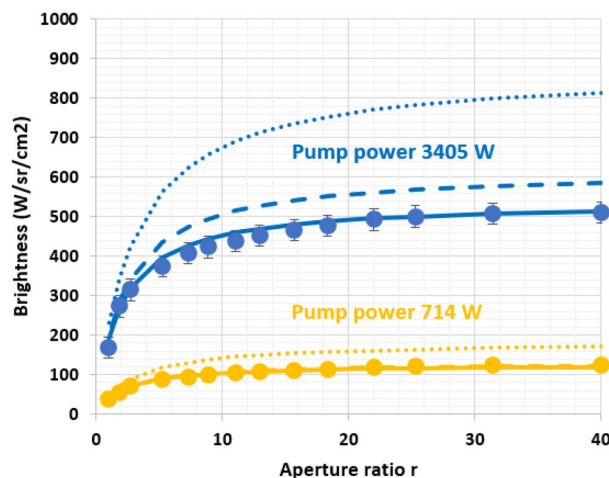


Figure 5. Brightness of the output surface for two different pump peak powers versus the aperture ratio. Dot are experimental values. Plains curves are simulations including excited state absorption (ESA) and self-absorption losses. Dashed curves are simulations without ESA but with self-absorption losses. Dot curves are simulations without ESA and without self-absorption.

$$P_{3D} = \frac{1}{r}(P_1 + P_2) \frac{T_{Fresnel}}{1 - e^{-2\alpha_{3D}d} \left[\frac{1 - T_{Fresnel}}{r} R + \left(1 - \frac{1}{r}\right) R^2 \right]} \quad (12)$$

where α_{3D} is the loss coefficient for propagation distances longer than $3l$, taking the light confinement in 3 dimensions in account. Compared to α_3 , self-absorption losses are adjusted at $\alpha_{inf} = 3 \times 10^{-3} \text{ cm}^{-1}$. α_{3D} is therefore: $\alpha_{3D} = \alpha_0 + \alpha_{inf} + \alpha_{ESA}$.

The other fitting parameters are adjusted as follows.

R is adjusted for the low pump power (714 W) where the effect of ESA is the lowest. We found $R = 92\%$. One must mention that this value is smaller than the mirror specifications (reflectivity of 99.3% at normal incidence). This smaller value can be related to an imperfect and partial reinjection of the recycled rays by the mirrors, due to the Lambertian emission of the concentrator. σ_{ESA} is adjusted for the high pump power (3405 W). We found $\sigma_{ESA} = 2.5 \times 10^{-17} \text{ cm}^2$. One must note that this value is just an estimation and may include other effects depending on the pump power.

Once again, simulations are carried out under different conditions: with or without ESA and self-absorption. In this 3D configuration, ESA has a much greater impact than in 2D configuration. This can be explained by the long distance travelled by rays before reaching the output. However, self-absorption appears to be much more important limitation.

Discussion

This work aims to better understand why a Ce:LYSO concentrator is less efficient than a Ce:YAG concentrator. Table 1 summarizes different parameters for these two crystals. Ce:LYSO has 3 parameters limiting the efficiency. By order of importance, the first one is the quantum efficiency, but this parameter could be improved by using Ce:LYSO crystals free of tetravalent cerium and specifically prepared to optimize quantum yield and not scintillation yield. The second one corresponds to self-absorption losses that limit the performance when the propagation distance increases, this parameter is intrinsic to the small Stokes shift of the Ce:LYSO in regard to the other garnet values (around 2600 cm^{-1} for Ce1 center and 3200 cm^{-1} respectively²⁶) and is related to the blue emission of the Ce:LYSO in regard to the green/yellow in the case of Ce:YAG. The third one is the excited state absorption, limiting the performance when the pump power increases.

Despite these limitations, Ce:LYSO can be compared to blue LEDs as shown in the Table 2. For a fair comparison, LED performance are measured in the same temporal regime ($40 \mu\text{s}$ 10 Hz) allowing to overdrive the LED and to obtain a peak power 2.6 times the specified power in CW. This value is limited by peak power saturation above a certain supply current.

In the 2D configuration, Ce:LYSO concentrator offers the power of approximately 30 blue LEDs in a bright line of $1 \times 22 \text{ mm}^2$ with a brightness higher than the one of a single LED. One has to note that a LED system emitting the same power (116 W) would emit on a line of $1 \text{ mm} \times 60 \text{ mm}$ assuming a state-of-the-art filling factor (50%) for the LEDs. This LED line would have a reduced brightness of $67.5 \text{ W cm}^{-2} \text{ sr}^{-1}$ caused by this filling factor. Therefore, with a brightness of $168 \text{ W cm}^{-2} \text{ sr}^{-1}$ and a power of 116 W, Ce:LYSO concentrator offers an unique combination clearly overcoming the performance of LED system.

In the 3D configuration, and for an aperture of 1 mm^2 , the Ce:LYSO concentrator delivers a power of 16 W with a brightness of $509 \text{ W cm}^{-2} \text{ sr}^{-1}$. To reach the same power with LEDs, a combination of 4 LEDs is then

Parameter description	Ce:LYSO	Blue LED
Dimensions (mm)	22 × 105 × 1	1 × 1
Pump pulse duration	40 μs	40 μs
Repetition rate	10 Hz	10 Hz
Output at full aperture $P_{2Dmirror}$	116 W	4.25 W
Brightness at full aperture	168 W cm ⁻² sr ⁻¹	135 W cm ⁻² sr ⁻¹
Output at squared 1mm ² aperture P_{3D}	16 W	4.25 W
Brightness at squared 1mm ² aperture	509 W cm ⁻² sr ⁻¹	135 W cm ⁻² sr ⁻¹

Table 2. Comparison of performance of Ce:LYSO and a state-of-the-art blue LED (Rubix line by Lumiled).

necessary. Assuming a filling factor of 50%, the square of 4 LED will measure 2.5×2.5 mm² and will have a brightness of 80 W cm⁻² sr⁻¹, six times smaller than the brightness of the Ce:LYSO output surface.

Figure 6 compares the spectral power density of the Ce:LYSO luminescent concentrator (square output surface of 1 mm²) to the ones of several LEDs operating in the same QCW regime (i.e. peak power 2.6 times the specified power in CW). It shows that the Ce:LYSO luminescent concentrator covers the performance of 4 LEDs (typically from 415 to 530 nm) with similar spectral power density. In comparison, spectral combining of LEDs would certainly use more than 4 LEDs to smooth spectrum and would barely overcome the brightness of one single LED regarding the complexity to propagate a LED beam in an optical device (large size and large divergence).

In conclusion, this work explores the potential of the scintillator material Ce:LYSO as a LED luminescent concentrator for solid-state lighting. An approach combining experiments and numerical simulations allows to explain the performance of Ce:LYSO. Indeed, low quantum efficiency, strong self-absorption and excited-state absorption affect the output power and the brightness, that are 2–4 times smaller than for the well-known Ce:YAG luminescent concentrator for the same pump power and for similar dimensions. However, Ce:LYSO luminescent concentrator combines power and brightness at a level that is not reachable by any geometrical assembly of blue LEDs. Besides, Ce:LYSO performance could be improved by using single crystals specially prepared for optimized quantum yield and not for scintillation properties where the presence of Ce⁴⁺ is preferred. This work also gives some outlook on the search of other Ce³⁺ doped blue emitting light concentrators. Better host than Ce:LYSO will have to combine higher quantum yield and Stokes shift typically greater than 3000 cm⁻¹.

Anyway, this new powerful and bright source could address applications in the field of LED illumination. As LEDs, a Ce:LYSO concentrator is an incoherent source providing speckle noise free-illumination with a larger acceptability than laser sources regarding light safety limits. So, Ce:LYSO luminescent concentrator could be seen as an interesting way to "magnify" the blue LED in terms of brightness and spectrum.

With such a broad spectrum in the blue, Ce:LYSO emission overlaps the absorption spectra of many dyes used in standard fluorescence microscopy. Preliminary investigations of confocal microscopy are reported with LEDs

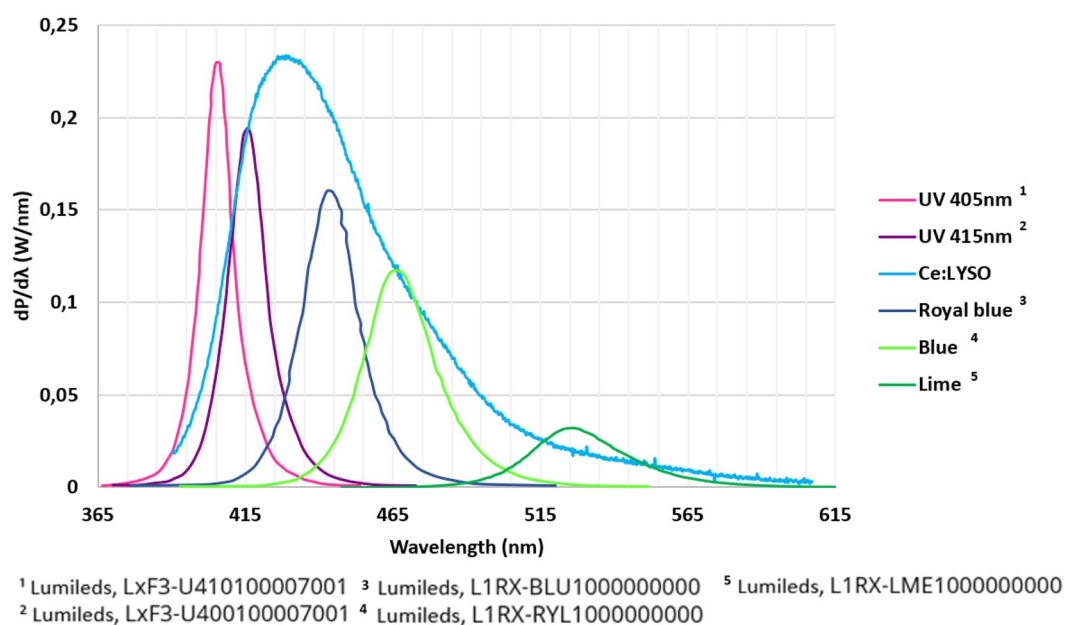


Figure 6. Spectral density of Ce:LYSO LC (square output surface of 1 mm²) and different LEDs.

of $1 \text{ W cm}^{-2} \text{ sr}^{-1}$ brightness²⁷. As the Ce:LYSO luminescent concentrator offers a brightness 500 times higher, it has a much larger potential for this application.

The short lifetime of Ce in LYSO (41 ns) makes the source able to emit 100 ns pulses. It can be used in photo-acoustic imaging where the duration of the illumination pulse limits the spatial "in depth" resolution: 100 ns corresponding to a spatial resolution of $150 \mu\text{m}$ ²⁸. Indeed, LED photo-acoustic imaging has a strong potential especially in the blue for high contrast in-vivo mapping of vasculature networks in biological tissue²⁹. However, the power density of LEDs limits the imaging depth (typically 1 cm). Ce:LYSO could therefore be a solution for exploring more clinical applications in the area of breast imaging and cardio-vascular medicine.

Data availability

Data underlying the results presented in this paper are not publicly available at this time but may be obtained from Lisa Lopez upon reasonable request.

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Author contributions

L. L. performed the experiment, performed the data analyses, wrote the manuscript; P. P. contributed to the conception of the study and helped the data analyses; P. L. and B. V. conducted the spectroscopy measurement and helped the understanding of the crystal; R. M. conducted the quantum yield measurement; F.D and P.G helped perform the analysis with constructive discussions; F. B. performed the conception of the study, helped the data analyses and supervised the writing of the manuscript. All authors reviewed the manuscript.

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Competing interests

The authors declare no competing interests.

Additional information

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